ELOISE A. MARAIS

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MAJOR RESEARCH AREA

I lead a research group that makes inventive use of atmospheric chemistry transport modeling and data from remote sensing platforms to interpret and determine the influence of humans on atmospheric composition, air quality and climate from Earth's surface to outer space.

EDUCATION

2008-2014 **Ph.D.**, Earth and Planetary Sciences, Harvard University, Cambridge MA, USA
Title: *Non-methane volatile organic compounds in Africa: A view from space*.
Supervisor: Prof. Daniel J. Jacob.
Funding: International Fulbright Science & Technology Award

2006-2008 **M.Sc** (research) in Chemistry, Rhodes University, South Africa, *distinction*.
Title: *Removal & photocatalysis of 4-nitrophenol using metallophthalocyanines.*
Supervisor: Prof. Tebello Nyokong

2004 **B.Sc. Hons** in Chemistry, University Kwa-Zulu Natal, South Africa, *cum laude*.

2001-2003 **B.Sc.** in Chemistry, University of Natal, South Africa, s*umma cum laude*.

ACADEMIC APPOINTMENTS AND PROFESSIONAL EXPERIENCE

University College London, London, UK

2024-present Professor of Atmospheric Chemistry and Air Quality, Department of Geography.
2020-2024 Associate Professor in Physical Geography, Department of Geography.

University of Pretoria, Pretoria, South Africa

2025-present Visiting Professor, Department of Geography, Geoinformatics and Meteorology.

University of Leicester, Leicester, UK

2018-2020 Associate Professor in Earth Observation, School of Physics and Astronomy.

University of Birmingham, Birmingham, UK

2016-2018 Tenure-track Research Fellow, School of Geography, Earth & Environ. Sciences.

Harvard University, Cambridge MA, USA

2014-2016 Postdoctoral Research Fellow in the Jacob group on a Schlumberger Faculty for the Future Fellowship, School and Engineering and Applied Sciences.

RESEARCH SUPPORT

Research grants awarded since 2017 totals £3.0 million.

In Progress

2027-2031 Leverhulme Research Project Grant, PI, *Seeing is Believing: Rocket plume pollution visualised with optical imagers*. **£430,000**.

2026-2027 Advanced Research + Invention Agency (ARIA) Scoping Our Planet Rolling Opportunity Seeds Call, PI, *Unlocking the potential to quantify spacecraft pollution from space-based broadband imagers*. **£425,000**.

Completed

2025-2026 UK Science and Technology Facilities Council (STFC) funding administered by UCL as a Knowledge Exchange Award, co-I, *Tracking pollution from rocket launches and object re-entries in partnership with the European Space Agency (ESA) Space Sustainability Team*. **£20,000**.

2019-2025 European Research Council (ERC) Starting Grant, PI, *UpTrop: Fundamental understanding of reactive nitrogen in the upper troposphere*. **£1.4 million**.

2021-2024 UK Natural Environment Research Council (NERC) Standard Grant, co-I, *Dry deposition processes of oxygenated volatile organic compounds*. **£160,000**.

2024 UK STFC funding administered by UCL as a Knowledge Exchange Award, PI, *Enhanced utility of Earth Observations for monitoring air quality using research-enabled gridding techniques*. **£22,000**.

2023 Higher Education Funding Council for England (HEFCE) Research Capital Infrastructure Funding (RCIF) administered through the UCL Research Capital Equipment Fund, PI, *Dynamics of ozone pollution and air pollution layer height in Central London*. **£83,000**.

2023 Research England's Higher Education Innovation Fund (HEIF) funding administered by UCL as an Innovation Fund Award, PI, *Engaging policymakers and industry to enhance the utility of a state-of-science air quality model*. **£13,600**.

2021-2023 Schmidt Family Foundation grant, co-I, *Health impacts and inequitable exposure to air pollution from the US oil and gas industry*. **£79,000**.

2018-2023 UK NERC Researcher In Residence Award, PI, *Launching urban air quality and green space monitoring into the 21st century*. **£50,000**.

2022 HEFCE RCIF administered through the UCL Research Capital Equipment Fund, PI, *First long-term MAX-DOAS instrument in London*. **£48,000**.

2019-2022 Copernicus Master's Accelerator Programme, PI, *Tool for Recording and Assessing the City Environment (TRACE)*. **£10,000**.

2020-2022 UK Department for Environment, Food and Rural Affairs (DEFRA) Air Quality Grant, PI, *Transboundary $PM_{2.5}$ sources in Leicester and Leicestershire*. **£124,000**.

2019-2020 UK DEFRA, co-I, *Applying Earth Observations (EO) to reduce uncertainties in regulatory emission inventories*. **£75,000**.

2019-2020 ESA, co-I, *Future EO passive optical missions for small satellites*. **£40,000**.

2017-2019 UK Department for International Development (DFID) East Africa Research Fund, co-I, *A systems approach to air pollution in East Africa*. **£20,000**.

2016-2018 University of Birmingham Independent Research Fellowship, PI. **£27,000**.

2016 Wallace Foundation, co-I, *Health effects of current and future fossil fuels*. **£8,000**.

2014-2015 Schlumberger Faculty for the Future Postdoctoral Fellowship, PI, *New regional inventory of diffuse and inefficient emissions for Africa*. **£59,000**.

2014-2015 South Africa National Research Foundation (NRF) Postdoctoral Fellowship for Research Abroad, PI. **£29,000**.

2011-2013 South Africa NRF, Graduate scholarship for study abroad. **£16,000**.

2011-2012 Harvard Center for the Environment Graduate Fellowship. **£12,000**.

2008-2011 International Fulbright Science & Technology PhD Scholarship. **£134,000**.

2004-2007 Multiple BSc Honours and Masters Scholarships, South Africa. **£10,000**.

PEER-REVIEWED PUBLICATIONS

Total citations: 7804**, h-index:** 39 (*Google Scholar*).

Students and postdocs supervised identified as: ** UG/MSc, * PhD, § Postdoc.

Archive of associated datasets: https://maraisresearchgroup.co.uk/datasets.html

2026

71. **E. A. Marais**, C. R. Barker§, A million data centre satellites will usher in a new era of space pollution, *ACS Earth and Space Chemistry*, submitted.

70. R. P. Horner*, **E. A. Marais**, L. T. Murray, Spatially and temporally resolved lightning NOx production rates obtained with space-based lightning detectors, *Environ. Sci.: Processes & Impacts*, doi:10.1039/D6EM00482B
69. K. Vohra§, **E. A. Marais**, R. P. Horner§, P. Sahoo, P. Mangaraj, A. Mishra, S. K. Sahu, Global health burden of ammonia emissions from fossil fuel derived synthetic nitrogen fertilizer use, *in review*, Earth's Future.
68. K. Vohra, W. J. Bloss, **E. A. Marais**, A. Mishra, P. Mangaraj, P. Sahoo, S. K. Sahu, C. Kumar, A. Chakraborty, Substantially underestimated health burden of Indian road transportation air pollution, *in review*, Nature Sustainability.
67. R. P. Horner*, **E. A. Marais**, R. M. Stauffer, D. E. Kollonige, A. M. Thompson, Global vertical profiles of tropospheric ozone (O_3) obtained by cloud-slicing TROPOMI, in review, *J. Geophys*.
66. **E. A. Marais**, N. Wei*, E. Y. P. Tan**, G. Lu, S. Keita, M. Naidoo, R. M. Garland, Urban and power plant NO_x emissions in Sub-Saharan Africa inferred from TROPOMI, *J. Geophys. Res.: Atmos.*, 131, doi:10.1029/2025JD045870.
65. C. R. Barker§, **E. A. Marais**, E. Y. P. Tan**, S. D. Eastham, G. S. Diskin, J. P. DiGangi, Y. Choi, A. Rollins, E. Waxman, T. P. Bui, C. Gatebe, J. Dean-Day, Radiative forcing and ozone depletion of a decade of satellite megaconstellation missions, *Earth's Future*, 14, doi:10.1029/2025EF007229.
64. M. Panagi, and 17 authors, including G. Lu§, **E. A. Marais**, Daily evolution of VOCs in Beijing: Chemistry, emissions, transport, and policy implications, Atmos. Poll. Res., 17, doi:10.1016/j.apr.2025.102783.

2025

63. K. Vohra§, **E. A. Marais**, P. Achakulwisut, S. Anenberg, C. Harkins, The health burden and racial-ethnic disparities of air pollution from the major oil and gas lifecycle stages in the United States, *Science Advances*, 11, eadu2241, doi:10.1126/sciadv.adu2241. ***Extensive media coverage:*** https://scienceadvances.altmetric.com/details/180522045.
62. N. Wei*, **E. A. Marais**, G. Lu§, R. G. Ryan§, B. Sauvage, Characterization of reactive nitrogen in the global upper troposphere using recent and historical commercial and research aircraft campaigns and GEOS-Chem, *Atmos. Chem. Phys.*, 25, 7925-7940, doi:10.5194/acp-25-7925-2025.
61. R. Dang, D. J. Jacob, H. Wang, C. R. Nowlan, G. González Abad, H. Chong, X. Liu, V. Shah, L. H. Yang, Y. J. Oak, **E. A. Marais**, R. P. Horner*, A. W. Rollins, J. H. Crawford, K. Li, H. Liao, High-resolution geostationary satellite observations of free tropospheric NO_2 over North America: Implications for lightning emissions, *PNAS*, 122, e2510535122, doi:10.1073/pnas.2510535122.
60. **E. A. Marais**, M. Van Damme, L. Clarisse, C. Wiedinmyer, K. Murphy, G. van der Werf, Subtropical southern Africa fire emissions of nitrogen oxides and ammonia obtained with satellite observations and GEOS-Chem, *Environ. Sci.: Atmos.*, 5, 906-920. *Invited*, doi:10.1039/d5ea00041f.
59. B. Opacka, T. Stavrakou, J.-F. Müller, I. De Smedt, J. van Geffen, **E. A. Marais**, R. P. Horner*, D. B. Millet, K. C. Wells, A. B. Guenther, Natural emissions of VOC and NO_x over Africa constrained by TROPOMI HCHO and NO_2 data using the MAGRITTEv1.1 model, *Atmos. Chem. Phys.*, 25, 2863–2894, doi:10.5194/acp-25-2863-2025.
58. T.-L. He, G.-M. Oomen, and 26 authors including **E. A. Marais**, Challenges and opportunities offered by geostationary space observations for air quality research and emission monitoring, *BAMS*, doi:10.1175/BAMS-D-23-0145.1.
57. G. Lu§, **E. A. Marais**, K. Vohra§, R. P. Horner*, D. Zhang, R. V. Martin, S. Guttikunda, Near-automated estimate of city nitrogen oxides emissions applied to South and Southeast Asia, *J. Geophys. Res.: Atmos.*, 130, e2024JD041000, doi:10.1029/2024JD041000.

2024

56. C. R. Barker[§], **E. A. Marais**, J. C. McDowell, Global 3D rocket launch and re-entry air pollutant and CO_2 emissions at the onset of the megaconstellation era, *Nature Scientific Data*., 11, 1-15, doi:10.1038/s41597-024-03910-z. **19 citations**.
55. R. P. Horner*, **E. A. Marais**, N. Wei*, R. G. Ryan[§], V. Shah, Vertical profiles of global tropospheric nitrogen dioxide (NO_2) obtained by cloud-slicing TROPOMI, *Atmos. Chem. Phys.*, 24, 13047–13064, doi:10.5194/acp-24-13047-2024.
54. K. Vohra[§], **E. A. Marais**, P. Achakulwisut, G. Lu[§], J. M. Kelly[§], C. Harkins, B. McDonald, Influence of oil and gas end-use on summertime particulate matter and ozone pollution in the eastern US, *Environ. Sci. & Techn*., 58, 19736–19747, doi:10.1021/acs.est.4c10032.

2023

53. **E. A. Marais**, J. M. Kelly[§], K. Vohra[§], Y. Li**, G. Lu[§], N. Hina, E. C. Rowe, Impact of legislated and best available emission control measures on UK particulate matter pollution, premature mortality, and nitrogen-sensitive habitats, *GeoHealth*, 7(10), e2023GH000910, doi:10.1029/2023GH000910. **12 citations**.
52. R. G. Ryan[§], **E. A. Marais**, E. Gershenson-Smith*, R. Ramsay, J.-P. Muller, J.-L. Tirpitz, U. Friess, Measurement Report: MAX-DOAS measurements characterise Central London ozone pollution episodes during 2022 heatwaves, *Atmos. Chem. Phys.*, 23, 7121–7139, doi:10.5194/acp-23-7121-2023. **25 citations**.
51. J. M. Kelly[§], **E. A. Marais**, G. Lu[§], J. Obszynska, M. Mace, J. White, R. J. Leigh, Diagnosing domestic and transboundary sources of fine particulate matter ($PM_{2.5}$) in UK cities using GEOS-Chem, *City & Environ. Interac*., 18, doi:10.1016/j.cacint.2023.100100. **24 citations**.

2022

50. R. G. Ryan[§], **E. A. Marais**, C. J. Balhatchet**, S. D. Eastham, Impact of rocket launch and space debris air pollutant emissions on stratospheric ozone and global climate, *Earth's Future*, 10, e2021EF002612, doi:10.1029/2021EF002612. ***Extensive media coverage:*** https://wiley.altmetric.com/details/129725006. ***Wiley certificate for most downloaded article in 2022***. **135 citations**.
49. K. Vohra*, **E. A. Marais**, W. J. Bloss, J. Schwartz, L. J. Mickley, M. Van Damme, L. Clarisse, P.-F. Coheur, Rapid rise in premature mortality due to anthropogenic air pollution in fast-growing tropical cities from 2005 to 2018, *Science Advances*, 8, doi:10.1126/sciadv.abm4435. **96 citations**.
48. **E. A. Marais**, O. Akker**, C. Wiedinmyer, Greenhouse gas and air pollutant emissions from power barges (powerships), *Roy. Soc. Chem. Environ. Sci.: Adv*., 1, 164-169, doi:10.1039/D1VA00049G.
47. R. Pope, R. Kelly, **E. A. Marais**, A. M. Graham, C. Wilson, J. J. Harrison, S. J. A. Moniz, M. Ghalaieny, S. R. Arnold, M. P. Chipperfield, Exploiting satellite measurements to reduce uncertainties in UK bottom-up NO_x emission estimates, *Atmos. Chem. Phys.*, 22, 4323–4338, doi:10.5194/acp-22-4323-2022.
46. A. Mazzeo, M. Burrow, A. Quinn, **E. A Marais**, A. Singh, D. Ng'ang'a, M. J. Gatari, F. D. Pope, Evaluation of WRF-CHIMERE coupled models for the simulation of $PM_{2.5}$ in large East African urban conurbations, *Atmos. Chem. Phys.*, 22, 10677–10701, doi:10.5194/acp-22-10677-2022.
45. B. Langford, E. House, A. Valach, C. N. Hewitt, P. Artaxo, M. P. Barkley, J. Brito, E. Carnell, B. Davison, A. R. MacKenzie, **E. A Marais**, M. J. Newland, A. R. Rickard, M. D. Shaw, A. M. Yáñez-Serrano, E. Nemitz, Seasonality of isoprene emissions and oxidation products above the remote Amazon, *Environ. Sci.: Atmos.*, doi:10.1039/D1EA00057H.

2021

44. **E. A. Marais**, A. Pandey[§], M. Van Damme, L. Clarisse, P.-F. Coheur, M. W. Shephard, K. E. Cady-Pereira, T. Misselbrook, L. Zhu, G. Luo, F. Yu, UK ammonia emissions estimated with

satellite observations and GEOS-Chem, *J. Geophys. Res.: Atmos.*, 126, doi:10.1029/2021JD035237, ***Wiley certificate for most downloaded article in 2021***. **43 citations**.

43. **E. A. Marais**, J. F. Roberts§, R. G. Ryan§, H. Eskes, K. F. Boersma, S. Choi, J. Joiner, N. Abuhassan, A. Redondas, M. Grutter, A. Cede, L. Gomez, M. Navarro-Comaset, New observations of upper tropospheric NO_2 from TROPOMI, *Atmos. Meas. Tech.*, 14, 2389-2408, doi:10.5194/amt-14-2389-2021. **48 citations**.
42. K. Vohra*, **E. A. Marais**, S. Suckra**, L. Kramer, W. J. Bloss, R. Sahu, A. Gaur, S. N. Tripathi, M. Van Damme, L. Clarisse, P.-F. Coheur, Long-term trends in air quality in major cities in the UK and India: A view from space, *Atmos. Chem. Phys.*, 21, 6275–6296, doi:10.5194/acp-21-6275-2021. **66 citations**.
41. K. Vohra*, A. Vodonos, J. Schwartz, **E. A. Marais**, M. P. Sulprizio, L. J. Mickley, Global mortality from outdoor fine particle pollution generated by fossil fuel combustion: Results from GEOS-Chem, *Environ. Res*., 195, 110754 doi:10.1016/j.envres.2021.110754. ***Extensive media coverage:*** https://www.altmetric.com/details/99822433, ***ISI Web of Science (WoS) Hot & Highly Cited Paper***. **1031 citations**.
40. D. A. Potts, **E. A. Marais**, H. Boesch, R. J. Pope, J. Lee, W. Drysdale, M. P. Chipperfield, B. Kerridge, R. Siddans, D. P. Moore, J. Remedios, Diagnosing air quality changes in the UK during the COVID-19 lockdown using TROPOMI and GEOS-Chem, *Environ. Res. Lett.*, 16, 054031, doi:10.1088/1748-9326/abde5d. **50 citations**.
39. B. Nault and 37 authors including **E. A. Marais**, Chemical transport models often underestimate inorganic aerosol acidity in remote regions of the atmosphere, *Nature Comm. Earth & Environ.*, 2, doi:10.1038/s43247-021-00164-0. **66 citations**.

2020

38. A. S. Bockarie*, **E A. Marais**, A. R. MacKenzie, Air pollution and climate forcing of the charcoal industry in Africa, *Environ. Sci. & Techn.*, 54, 13429–13438, doi:10.1021/acs.est.0c03754. **39 citations**.
37. P. deSouza, R. A. Kahn, J. A. Limbacher, **E. A. Marais**, F. Duarte, C. Ratti, Combining low-cost, surface-based aerosol monitors with size-resolved satellite data for air quality applications, *Atmos. Meas. Techn.*, 13, 5319–5334, doi:10.5194/amt-13-5319-2020.
36. E. E. McDuffie, S. J. Smith, P. O'Rourke, K. Tibrewal, C. Venkataraman, **E. A. Marais**, B. Zheng, M. Crippa, M. Brauer, R. V. Martin, A global anthropogenic emission inventory of atmospheric pollutants from sector- and fuel-specific sources (1970-2017): An application of the Community Emissions Data System (CEDS), *Earth Sys. Sci. Data*, 12, 3413-3442, doi:10.5194/essd-12-3413-2020. **541 citations**.
35. Y. Zheng, J. A. Thornton, N. L. Ng, H. Cao, D. K. Henze, E. E. McDuffie, W. Hu, J. L. Jimenez, **E. A. Marais**, E. Edgerton, J. Mao, Long-term observational constraints of organic aerosol dependence on inorganic species in the southeast US, *Atmos. Chem. Phys.*, 20, 13091–13107, doi:10.5194/acp-20-13091-2020.
34. S. J. Pai, C. L. Heald, J. R. Pierce, S. C. Farina, **E. A. Marais**, J. L. Jimenez, P. Campuzano-Jost, B. A. Nault, A. M. Middlebrook, H. Coe, J. E. Shilling, R. Bahreini, J. H. Dingle, K. Vu, An evaluation of global organic aerosol schemes using airborne observations, *Atmos. Chem. Phys.*, doi:10.5194/acp-20-2637-2020. **174 citations**.

2019

33. **E. A. Marais**, R. F. Silvern, A. Vodonos, E. Dupin**, A. S. Bockarie*, L. J. Mickley, J. Schwartz, Air quality and health impact of future fossil fuel use for electricity generation and transport in Africa, *Environ. Sci. & Techn.*, 53, 13524-13534, doi:10.1021/acs.est.9b04958. **98 citations**.

32. H. M. Worden, A. A. Bloom, J. R. Worden, Z. Jiang, **E. A. Marais**, T. Stavrakou, B. Gaubert, F. Lacey, New constraints on biogenic emissions using satellite-based estimates of carbon monoxide fluxes, *Atmos. Chem. Phys*., 19, 13569-13579, doi:10.5194/acp-19-13569-2019.
31. R. F. Silvern, D. J. Jacob, L. J. Mickley, M. P. Sulprizio, K. R. Travis, **E. A. Marais**, R. C. Cohen, J. L. Laughner, S. Choi, J. Joiner, L. N. Lamsal, Using satellite observations of tropospheric NO_2 columns to infer long-term trends in US NO_x emissions: The importance of accounting for the free tropospheric NO_2 background, *Atmos. Chem. Phys*., 19, 8863-8878, doi:10.5194/acp-19-8863-2019. **158 citations**.
30. D. S. Jo, A. Hodzic, L. K. Emmons, **E. A. Marais**, Z. Peng, B. A. Nault, W. Hu, P. Campuzano-Jost, J. L. Jimenez, A simplified parameterization of isoprene-epoxydiol-derived secondary organic aerosol (IEPOX-SOA) for global chemistry and climate models, *Geosci. Model Dev.*, 12, 2983-3000, doi:10.5194/gmd-12-2983-2019. **38 citations**.
29. J. Liao, T. F. Hanisco, G. M. Wolfe, J. St. Clair, J. L. Jimenez, P. Campuzano-Jost, B. A. Nault, A. Fried, **E. A. Marais**, G. González Abad, K. V. Chance, H. T. Jethva, T. B. Ryerson, C. Warneke, A. Wisthaler, Towards a satellite-in situ hybrid estimate for organic aerosol abundance, *Atmos. Chem. Phys.*, 19, 2765–2785, doi:10.5194/acp-19-2765-2019. **26 citations**.

2018

28. **E. A. Marais**, D. J. Jacob, S. Choi, J. Joiner, M. Belmonte-Rivas, R. C. Cohen, S. S. Beirle, L. T. Murray, L. Schiferl, V. Shah, L. Jaeglé, Nitrogen oxides in the global upper troposphere: Interpreting cloud-sliced NO_2 observations from the OMI satellite instrument, *Atmos. Chem. Phys.*, 18, 17017-17027, doi:10.5194/acp-18-17017-2018. **48 citations**.
27. C. L. Weagle and 40 authors, including **E. A. Marais**, Chemical sources of fine particulate matter: Interpretation of $PM_{2.5}$ chemical composition observed by SPARTAN using a global chemical transport model, *Environ. Sci. & Techn.*, 52, 11670-11681, doi:10.1021/acs.est.8b01658. **220 citations**.

2017

26. C. Li, R. V. Martin, A. Van Donkelaar, B. L. Boys, M. S. Hammer, J.-W. Xu, **E. A. Marais**, A. Reff, M. Strum, D. A. Ridley, M. Crippa, M. Brauer, Q. Zhang, Trends in chemical composition of global and regional population-weighted fine particulate matter over the recent 25 years, *Environ. Sci. & Techn.*, 51, 11185−11195, doi:10.1021/acs.est.7b02530. **113 citations**.
25. F. G. Lacey, **E. A. Marais**, D. K. Henze, C. J. Lee, A. van Donkelaar, R. V. Martin, M. P. Hannigan, C. Wiedinmyer, Improving present day and future estimates of anthropogenic sectoral emissions and the resulting air quality impacts in Africa, *Faraday Discuss.*, 200, 397-412, doi:10.1039/C7FD00011A. **39 citations**.
24. L. Zhu, L. J. Mickley, D. J. Jacob, **E. A. Marais**, J. Sheng, L. Hu, G. González Abad, K. V. Chance, Long-term (2005–2014) trends in formaldehyde (HCHO) columns across North America as seen by the OMI satellite instrument: Evidence of changing emissions of volatile organic compounds, *Geophys. Res. Lett.*, 44, 7079–7086, doi:10.1002/2017GL073859. **131 citations**.
23. K. E. Cady-Pereira, V. Payne, J. Neu, K. Bowman, K. Miyazaki, **E. A. Marais**, S. Kulawik, Z. Tzompa-Sosa, J. Hegarty, Seasonal and spatial changes in trace gases over megacities from Aura TES observations: Two case studies, *Atmos. Chem. Phys*., 17, 9379-9398, doi:10.5194/acp-17-9379-2017.
22. C. Chan Miller, D. J. Jacob, **E. A. Marais**, K. Yu, K. R. Travis, P. S. Kim, J. A. Fisher, L. Zhu, G. M. Wolfe, T. F. Hanisco, F. N. Keutsch, *et al.*, Glyoxal yield from isoprene oxidation and relation to formaldehyde: Chemical mechanism, constraints from SENEX aircraft observations, and interpretation of OMI satellite data, *Atmos. Chem. Phys.*, 17, 8725–8738, doi:10.5194/acp-17-8725-2017. **105 citations**.
21. H. M. Horowitz, D. J. Jacob, Y. Zhang, T. S. Dibble, F. Slemr, H. M. Amos, J. A. Schmidt, E. S. Corbitt, **E. A. Marais**, E. M. Sunderland, A new mechanism for atmospheric mercury redox

chemistry: Implications for the global mercury budget, *Atmos. Chem. Phys.*, 17, 6353-6371, doi:10.5194/acp-17-6353-2017. ***ISI WoS Hot & Highly Cited Paper***. **458 citations**.

20. **E. A. Marais**, D. J. Jacob, J. R. Turner, L. J. Mickley, Evidence of 1991-2013 decrease of biogenic secondary organic aerosol in response to SO_2 emission controls, *Environ. Res. Lett.*, 12, 054018, doi:10.1088/1748-9326/aa69c8. **57 citations**.
19. R. F. Silvern, D. J. Jacob, P. S. Kim, **E. A. Marais**, J. R. Turner, P. Campuzano-Jost, J. L. Jimenez, Inconsistency of ammonium-sulfate aerosol ratios with thermodynamic models in the eastern US: A possible role of organic aerosol, *Atmos. Chem. Phys.*, 17, 5107-5118, doi:10.5194/acp-17-5107-2017. **85 citations**.

2016

18. **E. A. Marais**, D. J. Jacob, J. L. Jimenez, P. Campuzano-Jost, D. A. Day, W. Hu, J. Krechmer, L. Zhu, P. S. Kim, C. Chan Miller, J. A. Fisher, K. R. Travis, K. Yu, T. F. Hanisco, G. M. Wolfe, H. L. Arkinson, H. O. T. Pye, K. D. Froyd, J. Liao, V. F. McNeill, Aqueous-phase mechanism for secondary organic aerosol formation from isoprene: Application to the Southeast United States and co-benefit of SO_2 emission controls, *Atmos. Chem. Phys.*, 16, 1603-1618, doi:10.5194/acp-16-1603-2016. ***ISI WoS Highly Cited Paper***. **342 citations**.
17. **E. A. Marais**, C. Wiedinmyer, Air quality impact of Diffuse and Inefficient Combustion Emissions in Africa (DICE-Africa), *Environ. Sci. & Techn.*, 50, 10739-10745, doi:10.1021/acs.est.6b02602. **170 citations**.
16. K. R. Travis, D. J. Jacob, J. A. Fisher, P. S. Kim, **E. A. Marais**, L. Zhu, K. Yu, C. Chan Miller, R. M. Yantosca, M. P. Sulprizio, *et al.*, Why do models overestimate surface ozone in the Southeast United States?, *Atmos. Chem. Phys.*, 16, 13561-13577, doi:10.5194/acp-16-13561-2016. ***ISI WoS Hot & Highly Cited Paper***. **457 citations**.
15. J. A. Fisher, D. J. Jacob, K. R. Travis, P. S. Kim, **E. A. Marais**, C. Chan Miller, K. Yu, L. Zhu, R. M. Yantosca, M. P. Sulprizio, *et al.*, Organic nitrate chemistry and its implications for nitrogen budgets in an isoprene- and monoterpene-rich atmosphere: Constraints from aircraft (SEAC^{4}RS) and ground-based (SOAS) observations in the Southeast US, *Atmos. Chem. Phys.*, 16, 5969-5991, doi:10.5194/acp-16-5969-2016. **250 citations**.
14. K. Yu, D. J. Jacob, J. A. Fisher, P. S. Kim, **E. A. Marais**, C. Chan Miller, K. R. Travis, L. Zhu, R. M. Yantosca, M. P. Sulprizio, R. C. Cohen, J. E. Dibb, A. Fried, T. Mikoviny, T. B. Ryerson, P. O. Wennberg, A. Wisthaler, Sensitivity to grid resolution in the ability of a chemical transport model to simulate observed oxidant chemistry under high-isoprene conditions, *Atmos. Chem. Phys.*, 16, 4369-4378, doi:10.5194/acp-16-4369-2016. **80 citations**.
13. B. Franco, **E. A. Marais**, B. Bovy, W. Bader, B. Lejeune, G. Roland, C. Servais, E. Mahieu, Diurnal cycle and multi-decadal trend of formaldehyde in the remote atmosphere near 46°N, *Atmos. Chem. Phys.*, 16, 4171-4189, doi:10.5194/acp-16-4171-2016. **38 citations**.

2015

12. B. Franco, F. Hendrick, M. V. Roozendael, J.-F. Müller, T. Stavrakou, **E. A. Marais**, B. Bovy, W. Bader, C. Fayt, C. Hermans, B. Lejeune, G. Pinardi, C. Servais, E. Mahieu, Retrievals of formaldehyde from ground-based FTIR and MAX-DOAS observations at the Jungfraujoch station and comparisons with GEOS-Chem and IMAGES model simulations, *Atmos. Meas. Tech.*, 8, 1733-1756, doi:10.5194/amt-8-1733-2015. **65 citations**.
11. **E. A. Marais**, K. V. Chance, A geostationary air quality monitoring platform for Africa, The Clean Air Journal, 25, 40-45, doi:10.17159/2410-972X/2015/v25n1a3.

2014

10. **E. A. Marais**, D. J. Jacob, K. Wecht, C. Lerot, L. Zhang, K. Yu, T. P. Kurosu, K. V. Chance, B. Sauvage, Anthropogenic emissions in Nigeria and implications for ozone air quality: A view from space, *Atmos. Environ*., 99, 32-40, doi:10.1016/j.atmosenv.2014.09.055. **115 citations**.

9. L. Zhu, D. J. Jacob, L. J. Mickley, **E. A. Marais**, D. S. Cohan, Y. Yoshida, B. N. Duncan, G. González Abad, K. V. Chance, Anthropogenic emissions of highly reactive volatile organic compounds in eastern Texas from oversampling of satellite (OMI) measurements of HCHO columns, *Environ. Res. Lett.*, 9, 114004, doi:10.1088/1748-9326/9/11/114004. **169 citations**.
8. C. R. Nowlan, R. V. Martin, S. Philip, L. N. Lamsal, N. A. Krotkov, **E. A. Marais**, S. Wang, Q. Zhang, Global dry deposition of nitrogen dioxide and sulfur dioxide inferred from space-based measurements, *Global Biogeochem. Cycles*, 28, 1025-1043, doi:10.1002/2014GB004805. **99 citations**.
7. **E. A. Marais**, D. J. Jacob, A. Guenther, K. V. Chance, T. P. Kurosu, J. G. Murphy, C. E. Reeves, and H. O. T. Pye, Improved model of isoprene emissions in Africa using Ozone Monitoring Instrument (OMI) satellite observations of formaldehyde: Implications for oxidants and particulate matter, *Atmos. Chem. Phys.*, 14, 7693-7703, doi:10.5194/acp-14-7693-2014. **71 citations**.
6. Q. Wang, D. J. Jacob, J. R. Spackman, A. E. Perring, J. P. Schwarz, N. Moteki, **E. A. Marais**, C. Ge, J. Wang, S. R. H. Barrett, Global budget and radiative forcing of black carbon aerosol: Constraints from pole-to-pole (HIPPO) observations across the Pacific, *J. Geophys. Res.: Atmos.*, 119, 195-206, doi:10.1002/2013JD020824. ***ISI WoS Highly Cited Paper***. **253 citations**.

2013

5. M. P. Barkley, I. De Smedt, M. Van Roozendael, T. P. Kurosu, K. V. Chance, A. Arneth, D. Hagberg, A. Guenther, F. Paulot, **E. A. Marais**, J. Mao, Top-down isoprene emissions over tropical South America inferred from SCIAMACHY and OMI formaldehyde columns, *J. Geophys. Res.: Atmos.*, 118, 6849-6868, doi:10.1002/jgrd.50552. **120 citations**.

2012

4. **E. A. Marais**, D. J. Jacob, T. P. Kurosu, K. Chance, J. G. Murphy, C. Reeves, G. Mills, S. Casadio, D. B. Millet, M. P. Barkley, F. Paulot, J. Mao, Isoprene emissions in Africa inferred from OMI observations of formaldehyde columns, *Atmos. Chem. Phys.,* 12, 6219-6235, doi:10.5194/acp-12-6219-2012. **211 citations**.

Prior to 2012

3. **E. A. Marais**, E. Antunes, T. Nyokong, Photocatalytic transformation of 4-nitrophenol in aqueous media using suspended, water-insoluble metallophthalocyanine complexes, *J. Coord. Chem.*, 61, 3727-3739, doi:10.1080/00958970802146056, 2008.
2. **E. A. Marais**, T. Nyokong, Adsorption of 4-nitrophenol onto Amberlite® IRA-900 modified with metallophthalocyanines, *J. Haz. Mat.*, 152, 293-203, doi:10.1016/j.jhazmat.2007.06.096, 2008. **151 citations**.
1. **E. A. Marais**, R. Klein, E. Antunes, T. Nyokong, Photocatalysis of 4-nitrophenol using zinc phthalocyanine complexes, *J. Molec. Cat. A: Chem.*, 261, 36-42, doi:10.1016/j.molcata.2006.07.055, 2007. **114 citations**.

REPORTS AND PERSPECTIVES

9. Lead contributing author, White Paper on "The Atmospheric impacts of re-entry and launch (AIRL): High-priority actions and recommendations", https://zenodo.org/records/20831147, 2026.
8. Issues Note on "Environmental impacts of space activity", UNEP, UNOOSA and Invited Experts, UN Office of the Chief Scientist, https://www.unep.org/resources/emerging-issues/safeguarding-space-environmental-issues-risks-and-responsibilities, 2025.
7. New opportunities for particulate matter measurements, UK Air Quality Experts Group (AQEG), https://tinyurl.com/bdd3r3x8 [pdf], 2024.
6. Cornwall space launch: Why the environmental cost of rocket launches is large even when they fail, *The Conversation*, https://theconversation.com/cornwall-space-launch-why-the-

environmental-cost-of-rocket-launches-is-large-even-when-they-fail-197567, 2023. ***4,822 reads, 5 comments***.

5. Air pollution in fast-growing African cities presents a risk of premature death, *The Conversation*, https://theconversation.com/air-pollution-in-fast-growing-african-cities-presents-a-risk-of-premature-death-183944, 2022. ***6,882 reads***.
4. Axiom launch: Why commercial space travel could be another giant leap for air pollution, *The Conversation*, https://theconversation.com/axiom-launch-why-commercial-space-travel-could-be-another-giant-leap-for-air-pollution-180990, 2022. ***41,982 reads, 20 comments***.
3. Ditching fossil fuels will have immediate health benefits for millions – world leaders must seize the chance, *The Conversation*, https://theconversation.com/ditching-fossil-fuels-has-immediate-health-benefits-for-millions-world-leaders-must-seize-the-chance-171015, 2021. ***9,552 reads, 2 comments***.
2. Space tourism: Rockets emit 100 times more CO_2 per passenger than flights – imagine a whole industry, *The Conversation*, https://theconversation.com/space-tourism-rockets-emit-100-times-more-co-per-passenger-than-flights-imagine-a-whole-industry-164601, 2021. ***121,943 reads, 53 comments***.
1. Household air pollution contribution to ambient air pollution in Ghana: Using available evidence to prioritize future action, Household and Ambient Air Pollution in Ghana Working Group, US Health Effects Institute (HEI), Communication 19, https://www.healtheffects.org/system/files/Comm19-HAP-Ghana.pdf, 2019.

INVITED PRESENTATIONS

Archive of all slides: http://maraisresearchgroup.co.uk/presentations.html

Conferences, Workshops and Society Events

Panelist, Royal Aeronautics Society Annual President's Conference, Space Sustainability Panel, London, UK, September 29, 2026.

Speaker, International Global Atmospheric Chemistry (IGAC) Conference, Heraklion, Crete, September 7-11, 2026.

Keynote, Royal Society of Chemistry Air Quality in the 21st Century Conference, London, UK, December 15, 2025.

Keynote, South African National Space Agency (SANSA) and University of Pretoria Satellite Observations of Air Quality Workshop, Pretoria, South Africa, October 21, 2025.

Keynote, 2nd International Conference on Chemical Weather and Chemical Climate, Ben Guerir, Morocco, October 14, 2025.

Expert, Joint UNEP/UNOOSA Outer Space Expert Interdisciplinary Meeting, UN Office at Vienna, Austria, May 22-23, 2025.

Speaker, Atmospheric Environmental Remote Sensing Society (AERSS) meeting, online, December 2, 2024.

Speaker, Royal Society UK-South Africa Bilateral Frontiers of Science meeting, Johannesburg, South Africa, September 24-27, 2024.

Speaker, 2nd International Space Science Institute (ISSI) Workshop on Geostationary Satellites, Beijing, China, May 28, 2024.

Speaker, Royal Society UK-China Bilateral Air Quality and Climate Change Workshop, London, UK, May 22-23, 2024.

Speaker, 1st ISSI Workshop on Geostationary Satellites, Bern, Switzerland, March 20-23, 2023.

Speaker, Follow-up Workshop on a Pilot Design for Air Quality in Africa, Kigali, Rwanda, January 17-18, 2023.

Speaker, Royal Society of Chemistry Desktop Seminar series, online, June 1, 2022.

Keynote, ERC Health, Labor and Environmental Regulation in Post-Industrial Europe (HEAL) Project Workshop, University of Mannheim, Mannheim, Germany, July 7, 2022.

Speaker, Investigation of Air Pollution Standing Conference, online, December 1, 2021.

Speaker, DEFRA Earth Observation Centre of Excellence Workshop, online, September 21, 2021.

Speaker, Townhall on African Greenhouse Gas Emissions and Air Quality, organized by the Manchester Environmental Research Institute, online, March 30, 2021.

Speaker, National Center for Atmospheric Research (NCAR) Workshop on Advancing Air Quality and Carbon Science in Africa, March 11, 2021.

Speaker, US HEI webinar on Understanding the Sources of Outdoor Air Pollution in Sub-Saharan Africa, August 21, 2019.

Speaker, Royal Meteorological Society National Meeting on Air Pollution in Megacities, University of Leeds, Leeds, UK, March 20, 2019.

Lecturer, British Council Air Quality Capacity Building Workshop for PhD students in East Africa, University of Nairobi, Nairobi, Kenya, February 11, 2019.

Speaker and Panelist, Air Pollution Extremes Workshop, Columbia University, New York NY, November 1-2, 2018.

Lecturer, NASA/COSPAR Capacity Building Workshop for PhD students across West Africa, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana, June 12-23, 2017.

Academic and Research Institute Seminars

Department of Meteorology seminar series, University of Reading, Reading, UK, November 10, 2025.

Royal Belgian Institute for Space Aeronomy (BIRA) seminar series, Brussels, Belgium, October 23, 2025.

Institute for Atmospheric and Climate Science seminar series, ETH, Zurich, October 2, 2025.

Wolfson Atmospheric Chemistry Laboratories seminar series, University of York, York, UK, November 17, 2023.

Global Change Seminar Series, University of Edinburgh, Edinburgh, UK, November 30, 2022.

Harvard Climate Tea, Harvard University, Cambridge MA, November 9, 2022.

Lunch Hour Lecture Series, UCL, London, UK, February 1, 2022.

Global Air Quality Conversation seminar series, University of Peking, China, September 23, 2022.

Atmospheric and Environmental Chemistry seminar series, Harvard University, Cambridge MA, October 1, 2021.

Air Pollution and Atmospheric Chemistry seminar series, University of Birmingham, Birmingham, UK, April 21, 2021.

Program in Atmospheres, Oceans and Climate seminar series, MIT, Cambridge MA, April 5, 2021.

Institute for Climate & Atmospheric Science, University of Leeds, Leeds, UK, March 16, 2021.

Centre for Atmospheric Science seminar series, University of Cambridge, Cambridge, UK, November 30, 2020.

Royal Netherlands Meteorological Institute (KNMI) seminar series, De Bilt, The Netherlands, November 12, 2019.

Environment, Earth and Ecosystem Sciences seminar series, Open University, Milton Keynes, UK, February 19, 2019.

Special Atmospheric Chemistry Seminar, Harvard University, Cambridge MA, November 8, 2018.

Council for Scientific and Industrial Research, Pretoria, South Africa, August 24, 2018.

Centre for Atmospheric Science seminar series, University of Cambridge, Cambridge, UK, November 20, 2017.

School of Earth and Environmental Sciences seminar series, University of St Andrews, St Andrews, UK, November 7, 2017.

UK Centre for Ecology and Hydrology, Edinburgh, UK, October 15, 2017.

Geosciences seminar series, University of Edinburgh, Edinburgh, UK, May 17, 2017.

School of Geography, Earth and Environmental Sciences seminar series, University of Birmingham, Birmingham, UK, July 8, 2016.

Earth, Atmospheric and Planetary Sciences seminar series, MIT, Cambridge MA, February 19, 2016.

Texas A&M Atmospheric Sciences seminar series, College Station TX, February 11, 2016.

Chemistry seminar series, Rhodes University, South Africa, November 16, 2015.

Civil and Environmental Engineering seminar series, Georgia Institute of Technology, Atlanta GA, October 28, 2015.

School of Environmental Sciences seminar series, University of East Anglia, East Anglia, UK, May 20, 2015.

Department of Chemistry and National Centre for Atmospheric Science seminar series, University of York, York, UK, May 26, 2015.

School of Chemistry seminar series, University of Leeds, Leeds, UK, May 27, 2015.

School of Earth, Atmospheric and Environmental Sciences seminar series, University of Manchester, Manchester, UK, May 28, 2015.

Cooperative Institute for Research in Environmental Sciences seminar series, University of Colorado, Boulder CO, February 11, 2015.

Atmospheric Chemistry Observations and Modeling seminar series, NCAR, Boulder CO, February 9, 2015.

Unit of Environmental Sciences and Management, North-West University, Potchefstroom, South Africa, March 28, 2014.

Stakeholders and End Users

Panelist, 5th European Union Clean Air Forum Earth Observations and Clean Air Panel, Bonn, Germany, December 1, 2025.

Speaker, *Health burdens and racial-ethnic disparities of air pollution from US oil and gas lifecycle stages*, Sierra Club, online, November 13, 2025.

Speaker, ESA Essential Climate Variables (ECV) Climate Change Initiative (CCI) Users Workshop, online, June 11, 2025.

Speaker, UK DEFRA Deep-Dive Series, Episode 6: Detection of Ammonia from Agriculture, online, February 27, 2025.

Speaker, DEFRA Emissions Monitoring Deep-Dive Symposium, online, November 14, 2024.

Speaker, Institution of Environmental Sciences (IES) webinar series, November 22, 2022.

Speaker, Great Ormond Street Hospital (GOSH) lunchtime seminar, online, May 18, 2021.

Speaker, Joint European Environmental Information and Observation Network (EIONET) and Task Force on Emission Inventories and Projections (TFEIP) meeting, online, May 6, 2021.

Speaker, Oxford Air Quality Meeting, Keble College, University of Oxford, Oxford, UK, January 10, 2020.

Speaker, UK Connected Places Catapult Innovation Hub Lunchtime Talk Series, London, UK September 10, 2019.

PROFESSIONAL ACTIVITIES

National and International Expert Committees

2026-present Member of the UK AQEG providing air quality advice to the UK government.

2026-2027 Member of the Air Quality Expert Review Group (AQERG) for the Heathrow Expansion Project.

2025-present Steering Committee for the Belgian Science Policy Office (BELSPO) funded Belgian Ammonia assessed using innovative Multiscale Measurements and Modelling (BEAM) project led by Université Libre de Bruxelles (ULB), Brussels, Belgium.

2024-present International GEOS-Chem Model Co-lead.

2017-present International GEOS-Chem Model Steering Committee, Co-Chair of the Emissions and Deposition Working Group.

2024-present Atmospheric Impacts of Spacecraft Re-entry and Launch (AIRL) Working Group.

2022-2023 UK AQEG invited ad-hoc member contributing niche expertise in Earth Observations of particulate matter air pollution.

2020-2023 US HEI Global Health Oversight Committee.

2018 US HEI Household and Ambient Air Pollution in Ghana Working Group.

Organizer, Chair and Convener of Conferences, Workshops and Seminar Series

2026 Session chair, 12th International GEOS-Chem User's Meeting, Washington University in St Louis, St Louis MO, June 9-12.

2026 Session convener, European Geophysical Union (EGU) General Assembly, Vienna, Austria, May 3-8.

2026 Co-organizer, ISSI Workshop on the Atmospheric Impacts of Spacecraft Re-entry, Bern, Switzerland, 2026, proposal under evaluation.

2025 Co-organizer, 1st UK Atmospheric Chemistry Conference, York, UK, September 9-10.

2025 Organizer, 3rd GEOS-Chem Europe Regional Meeting, UCL, London, UK, August 18-20.

2025 Session convener, EGU General Assembly, Vienna, Austria, April 28 to May 2.

2024 Session co-chair, American Geophysical Union (AGU) Annual Meeting, Washington DC, December 9-13.

2024 Session chair, 11th International GEOS-Chem User's Meeting, Washington University in St Louis, St Louis MO, June 11-14.

2024 Session co-convener, EGU General Assembly, Vienna, Austria, April 15-19.

2023 Co-organizer and session co-chair, HEI Workshop on Air Quality in East Africa, Nairobi, Kenya, March 29-31.

2023 Organizer, 2nd GEOS-Chem Europe Regional Meeting, UCL, London, UK, August 14-16.

2023 Session co-chair, AGU Annual Meeting, San Francisco CA, December 11-15.

2022 Session co-chair, HEI Workshop on Health Applications for Satellite-Derived Air Quality, online, April 29.

2022 Session chair, 10th International GEOS-Chem User's Meeting, Washington University in St Louis, St Louis MO, June 7-10.

2021 Session co-chair, HEI Annual Conference, online, April 20.

2020 Session co-chair, AGU Annual Meeting, online, December 7-11.

2020 Co-organizer, 1st GEOS-Chem Europe Regional Meeting, online, September 1-2.

2019 Session chair, 9th International GEOS-Chem User's Meeting, Harvard University, Cambridge MA, May 6-19.

2018 Session co-chair, AGU Annual Meeting, Washington DC, December 10-14.

2018 Session co-chair, UK National Centre for Earth Observation conference, September 4-7.

2017 Session co-chair, AGU Annual Meeting, New Orleans LA, December 11-15.

2017 Organizer, Air Pollution and Atmospheric Chemistry seminar series, University of Birmingham, Birmingham, UK.

2015 Organizer, Atmospheric Chemistry seminar series, Harvard University, Cambridge MA.

Grant Review

2025 UK Natural Environment Research Council (NERC) Future Leaders Fellow review
2024 UK NERC Highlights Topics review
2024 UK NERC Pushing the Frontiers review panel
2022 BELSPO BRAIN-be 2.0 Research Programme review
2022 UK Engineering and Physical Sciences Research Council (EPSRC) review panel
2020 Canada Foundation for Innovation's (CFI) Innovation Fund review panel
2020 UK NERC Strategic Priorities Fund review panel
2020 BELSPO BRAIN-be 2.0 Research Programme review
2019 US EPA STAR Grant reviewer and review panel
2019 Irish Research Council Government of Ireland Postdoctoral Fellowship Scheme reviewer
2017 UK NERC National Capability: Official Development Assistance call review
2017 US National Oceanic and Atmospheric Association (NOAA) Atmospheric Chemistry, Carbon Cycle and Climate (AC4) Program review
2016 European Research Council (ERC) Advanced Grant review
2015 NOAA AC4 Program review
2015 BELSPO STEREO Research Programme review
2013 NASA Research Opportunities in Earth and Space Science Carbon Cycle Science call review

Peer Review

Frequent reviewer for general readership and discipline-specific science journals:

Proceedings of the National Academy of Sciences; Science Advances; Nature Communications; Scientific Reports; Environmental Science & Technology (ES&T); ES&T Letters; Atmospheric Environment; Atmospheric Chemistry and Physics; Atmospheric Measurement Techniques; Journal of Air & Waste Management; Geoscientific Model Development; Journal of Geophysical Research: Atmospheres; Aerosol & Air Quality Research; Environment International; Science of the Total Environment; GeoHealth; NPJ Climate & Atmospheric Science, Royal Society of Chemistry's Environment Science: Atmospheres, Elementa: Science of the Anthropocene; Earth's Future.

Assessment Reports Review

2022 European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT) Atmospheric Composition Monitoring Satellite Application Facilities (AC SAF) Infrared Atmospheric Sounding Interferometer (IASI) ammonia (NH_3) data products
2018 IGAC project Tropospheric Ozone Assessment Report (TOAR) on Present-day distribution and trends of tropospheric ozone relevant to climate and global atmospheric chemistry model evaluation by A. Gaudel *et al.*

Consultancy

2025-2026 Consultant to Industrial Economics, Incorporated (IEc) for the C40 Cities funded "Air Quality Baseline Emissions, Health and Equity Impact Assessment in Freetown" study.
2025 Follow-on expert critique of Climate Change and Environment Impact Assessments (CCIA, EIA) resubmitted by Tetra4 following my 2023 critique, commissioned by lawyers at South African firms Natural Justice and the Centre for Environmental Rights.
2023 Expert critique of CCIA and EIA submitted by Tetra4 for proposed natural gas production at a UNESCO World Heritage Site in South Africa, commissioned by lawyers at Natural Justice and the Centre for Environmental Rights.
2023 Expert critique of CCIA and EIA submitted by Total for proposed offshore natural gas production along the Southern Cape Coast of South Africa, commissioned by lawyers at Earthjustice (US) and Natural Justice (South Africa).

Arbitration

2026 Emissions expert report and witness at arbitration for global legal firm Herbert Smith Freehills Kramer.

Institutional Service

2025-present Institutional Space Domain Stakeholder Group, Faculty representative, UCL.
2023-2026 Departmental Graduate Tutor, non-academic support for 100+ PhD students in Human and Physical Geography, Department of Geography, UCL.
2023-2026 Departmental Management Committee, Department of Geography, UCL.
2023-2026 Departmental Education Committee, Department of Geography, UCL.
2023-2026 Postgraduate Taught Examine Board, Department of Geography, UCL.
2023-2026 Faculty Research Degrees Committee, Social and Historical Sciences, UCL.
2023 Hiring Committee, Lecturer in Earth Observation, Department of Geography, UCL.
2019 Coordinator, First Leicester chapter of the International Pint of Science Festival, University of Leicester, Leicester, UK.

RESEARCH SUPERVISION

Professional Development

Advancing Principal Investigators, multiday intensive research team leadership training, UCL, 2022.

Current Supervision

2026- Yiyuan Jia (**PhD**) *Drivers and environmental impacts of trends in anthropogenic ammonia pollution in Central Africa inferred with satellite observations*. Self funded.
2024- Huilin Zhan (**PhD**) *Biogenic small acids from multiphase processing of isoprene epoxydiols*. Self funded.
2023- Connor Barker (**postdoc**) *Influence of megaconstellations on climate and air quality; Dry deposition of oxygenated VOCs*, funded by ARIA, the ERC, NERC, and UCL.
2022- Terrence Sepuru (**PhD**), *Quantifying air pollutant emissions over the intensely industrialized Highveld region in South Africa*, funded by the South African NRF.

Past Supervision

2022-2026 Eleanor Gershenson-Smith (**PhD**). Now a postdoc at UCL.
2021-2025 Rebekah Horner (**PhD**). Now a research scientist at Ember (think tank), London.
2025 Eric Tan (**UG**). Now a Master's student at UCL.
2019-2025 Nana Wei (**PhD**). Now a postdoc at University of Birmingham.
2017-2024 Karn Vohra (**PhD, postdoc**). Now a postdoc at University of Birmingham.
2017-2023 Gongda Lu (**MSc, PhD, postdoc**). Now a data scientist at the Centre for Research on Energy and Clean Air (CREA) (independent research organization).
2020-2022 Robert Ryan (**postdoc**). Now a postdoc at University of Melbourne, Australia.
2021-2022 Jamie Kelly (**postdoc**). Now a research scientist at CREA.
2022 Yifan Li (**UG**). Now a PhD student at Princeton University.
2022 Marco Barnfield (**UG**). Now a PhD student at UCL.
2021-2022 Kavitha Mottungan (**postdoc**). Now a senior researcher at the UK National Physical Laboratory, London.
2021 Orianna Akker (**UG**). Now a data analyst at Lloyds.
2020 Alok Pandey (**postdoc**). Now a lecturer at University of Delhi.
2017-2021 Alfred Bockarie (**PhD**). Now a lecturer at University of Njala, Sierra Leone.
2020 Chloe Balhatchet (**UG**). Now a PhD student at University of Cambridge.
2019 Junju Ng (**MSc**). Now a Meteorologist at the Singapore National Environment Agency.

2019 Eleanor Dupin (**UG**). Now a Project Manager Officer Analyst at MIGSO-PCUBED.

2018 Shannen Suckra (**MSc**). Now a research scientist at the National Environment and Planning Agency in Jamaica.

2018 Isobel Ward (**MSc**). Now at AECOM.

PhD Examination

2026 M. Zara, University of Wageningen, The Netherlands.
2025 B.A.H. Gutierrez, University of Toronto, Canada.
2025 C. Poraicu, Université Libre de Bruxelles, Belgium.
2023 L. Fakes, University of York, UK.
2023 R. Abeed, Laboratoire Atmosphères, Observations Spatiales (LATMOS), France.
2022 C. Mogno, University of Edinburgh, UK.
2022 I. Riádigos Sánchez, University of Santiago de Compostela, Spain.
2022 Y. Liu, University of Helsinki, Finland.
2022 C. K. Segakweng, North-West University, South Africa.
2021 D. Oyarzun, Department of Geography, UCL.
2020 A. de Lange, University of Pretoria, South Africa.
2019 L. Gonzalez Alonso, University of Sheffield, UK.

TEACHING

Training and Accreditation

2020 Fellow of the UK Higher Education Academy (FHEA).

2010 Harvard Bok Center Teaching Assistant Training, Harvard University, Cambridge MA.

2009 Scientists Teaching Science, graduate course on effective science teaching, Harvard-Smithsonian Center for Astrophysics, Cambridge MA.

Current

2026 Lecturer, GEOG0186: London Air Quality, 1st year module.

2026 Invited Lecture, CEGE0015: Urban Ozone Pollution, Master's module on Environmental Systems in Civil Engineering.

2025-2026 Lecturer, GEOG0109: Models in Environmental Science, Master's module.

2025-2026 Lecturer, GEOG0040: Principles and Practice of Remote Sensing, Master's module.

2020-2026 Supervisor, GEOG0042: Independent Study Project, 3rd year module.

2021-2026 Supervisor, GEOG0105: Research Project and Dissertation, Master's module.

Past

2026 Convener and Lecturer, GEOG0187: Quantitative Skills, 1st year module.

2024 Lecturer, BIOS0045: Foundations of Nature and Climate-Friendly Urban Design, Master's module, Department of Biosciences, UCL.

2022-2023 Convener and Lecturer, GEOG0170: Environmental Consequences of Human Activity, 3rd year module, Department of Geography, UCL.

2020-2023 Lecturer, GEOG0005: Understanding Our Planet, 1st year module, Department of Geography, UCL.

2019-2020 Lecturer, Waves and Quanta, 1st year module, Department of Physics and Astronomy, University of Leicester.

2019 Course Developer, Master's in Satellite Data Science, Department of Physics and Astronomy, University of Leicester.

2017 Lecturer, Environmental Protection, 3rd year module, School of Geography, Earth and Environmental Science, University of Birmingham.

2009-2013 Postgraduate Teaching Assistant for multiple courses (Energy Technology, Introduction to Environmental Science, Environmental Science & Technology, and Atmospheric Chemistry), Harvard University.

SELECTED MEDIA ENGAGEMENT

Extensive record of media engagement: https://maraisresearchgroup.co.uk/media_coverage.html

Environmental impacts of spacecraft launches and re-entries:

The Independent, Satellite launches are filling up the atmosphere with pollution, study warns, https://www.independent.co.uk/space/launch-pollution-satellite-study-new-b2977364.html, May 15, 2026.

NPR's All Things Considered, https://www.npr.org/2026/05/17/nx-s1-5821086/what-scientists-are-learning-about-geoengineering-from-satellite-pollution, May 17, 2026.

IFLScience, Pollution From Megaconstellation Launches Is Changing The Atmosphere, Causing "A Climate Effect That's A Little Bit Like Geoengineering", https://www.iflscience.com/pollution-from-megaconstellation-launches-is-changing-the-atmosphere-causing-a-climate-effect-thats-a-little-bit-like-geoengineering-83506, May 14, 2026.

Yale Environment 360, Amid a satellite boom, scientists warn of emissions risks, https://undark.org/2025/12/15/emissions-satellite-surge/, December 15, 2025.

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